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A Multi-State Analysis of Backyard Pool-Barrier Condition Between Certifications, Against a Register That Reports Certificates Current

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Abstract. A backyard pool barrier is certified at an inspection and recorded on a council register for three years, and the register's published compliance rate is the share of registered pools whose certificate is current. We treat that arrangement as a measurement problem: the certificate describes a barrier's state at one instant, while the quantity it is read as describing is the state of barriers in general. With the consent of 312 households holding a current certificate, we conducted 1,058 unannounced re-inspections at randomised intervals over 76 weeks, applying the certifier's own four-criterion checklist, and fitted a reversible three-state Markov model to the resulting panel. At a randomly chosen moment, 61.4% of certified barriers met every criterion (95% CI 58.4 to 64.3), against a register figure of 98.1% for the same population. The fitted mean sojourn time in the fully compliant state is 14.6 weeks (95% CI 11.9 to 18.2), and state occupancy is within one percentage point of its stationary distribution by week 26 of a 156-week certificate. Most departures are small and reversible: an object left inside the non-climbable zone accounts for 57.9% of single-defect observations, and 84.3% of single defects observed at one visit are absent at the next. Our results suggest the certificate may carry information for a minority of its validity period, though the direction of the estimate is stable across every specification we examined.

Introduction

A pool barrier is one of the few safety controls whose entire function is visible from the back step. It either encloses the water on four sides, or it does not; the gate either swings shut and latches from wherever it was left, or it does not; the space beside the fence either holds a chair, a pot, or a wheeled bin, or it is clear. Nothing about assessing it requires instruments, and the assessment takes a certifier about twenty minutes.

What the certifier produces, however, is not an assessment of the barrier. It is an assessment of the barrier at 10:40 on a Tuesday, which is then recorded on a register, given a three-year validity, and aggregated quarterly into a figure described as pool safety compliance. The council whose register we study reported that figure as 98.1% across 6,140 registered private pools in the quarter our fieldwork began. The number is correct. It counts certificates.

The gap between those two readings is not a failure of the register; it is a property of any scheme that certifies a state and reports a stock. It becomes an empirical question only when someone goes and looks in between, which is rarely done, because the households have already passed and the certifiers have already moved on. Barriers are among the interventions with the most consistent supporting evidence in the drowning-prevention literature [1], [2], and the drowning record repeatedly implicates barriers that were absent, incomplete, or defective at the moment they were needed [3], [4]. The interval between certification and need is therefore the interval of interest, and it is the one the register does not observe.

We contribute the following, each qualified as the design allows:

- A panel of unannounced re-inspections of certified backyard pool barriers, conducted with the certifier's own checklist, which yields a point prevalence of full compliance materially below the register's reported rate.
- A reversible multi-state model of barrier condition that estimates how long a barrier remains in the state its certificate names, and how quickly the population forgets when it was certified.
- A direct statement of the two estimands side by side, which we offer as a measurement result rather than as a finding about any particular household or certifier.

Background

Pool fencing is unusual among home safety measures in having a settled evidence base: isolation fencing that separates the pool from the house outperforms perimeter fencing that merely encloses the property [2], and reviews of drowning-prevention interventions place barrier legislation among the measures with supporting evidence [1], against a mortality burden that remains substantial [5]. Provision schemes that supply equipment rather than advice alone increase the presence of home safety equipment [6]; what happens to that equipment afterwards is less studied.

The regulatory literature has treated the general problem at length. Inspection regimes change behaviour and outcomes when they are conducted, including when allocation is randomised [7], and the effectiveness of monitoring and enforcement is by now reviewed rather than debated [8]. Public disclosure of an inspection result shifts both the result and the conduct behind it [9]. The complementary finding is equally settled: an indicator that becomes the reported quantity begins to be managed as such [10], [11], and public measures reshape the practices they claim to describe [12]. None of this literature turns on whether the indicator is arithmetically correct. Ours does not either.

Prior work at this institution has twice found the same shape in adjacent apparatus. A deployment of self-testing smoke alarms across 214 rental properties found that a managing agent's acknowledgment of a weekly self-test fault held high and uniform across fault types while actual dispatch did not [13]; and an audit of six years of break-glass activation logs across 214 call points found that 96.4% of logged activations were scheduled tests [14]. In both cases the record was accurate and described the apparatus's own routine rather than the events it existed to capture.

The statistical machinery is standard. Where a process is observed intermittently and its states are reversible, the panel-observed Markov model is the estimator of record [15], [16], within the broader multi-state framework for event-history data [17], [18], [19], and its assumptions are checkable [20].

Method

Setting. One metropolitan council maintains a register of 6,140 private swimming pools. A barrier is inspected by an accredited certifier; if it meets the standard, a certificate of compliance is issued and recorded, and the record stands for three years. The council's quarterly report publishes the share of registered pools holding a current certificate. The three-year interval predates the register's move off paper and has not been revised since.

Sample. From the 4,206 pools certified within the preceding two years, we drew a stratified random sample of 900 households and invited them to a study of barrier condition, with 312 consenting (34.7%). Consent covered up to five unannounced exterior inspections over 76 weeks, at intervals drawn from a schedule fixed before recruitment, with no notice of any individual visit and no feedback until the study closed. Households were told the interval distribution but not their own draws. We completed 1,058 inspections, a mean of 3.39 per dwelling.

Instrument. Inspectors applied the four criteria the council's certifiers apply: the gate self-closes and self-latches from any open position; no climbable object stands within the 900 mm non-climbable zone; no gap exceeding 100 mm appears below or between barrier members; and the barrier height and latch position meet the standard. Two inspectors independently scored a 12% subsample, agreeing on the overall verdict in 96.7% of cases.

States. Inspections were classified into three states: S_1, all four criteria met; S_2, exactly one criterion failed and restorable without tools or materials; S_3, two or more criteria failed, or one failure requiring repair. The chain is reversible between adjacent states, since the commonest departure from S_1 is an object that can be moved back.

Estimation. With generator Q and states observed only at inspection times, the interval likelihood uses the transition probability matrix

$$P(t) = \exp(Qt), \quad [P(t)]_{rs} = \Pr(X_{u+t} = s \mid X_u = r),$$

maximised over the panel with household random effects on the $S_1 \to S_2$ intensity. The quantities we report are the stationary distribution π satisfying $\pi Q = 0$, the point prevalence of S_1 at a uniformly random moment in a certificate's life, and the mean sojourn time $-1/q_{rr}$ in each state. Intervals are non-parametric bootstrap percentiles over 2,000 resamples of households. Six specifications were pre-specified: the primary three-state model, a two-state collapse, a four-state split of S_3, a hidden Markov variant allowing inspector misclassification, a restriction to households with three or more completed visits, and a version reweighted to the register's pool-age distribution.

Results

Across 1,058 unannounced inspections, 61.4% of certified barriers met all four criteria (95% CI 58.4 to 64.3), 27.6% carried a single remediable defect, and 11.0% carried two or more or a structural failure. The register's published figure for the same population over the same period was 98.1%, and every dwelling in our sample was inside it throughout: no certificate lapsed during the study.

Composition by time since certification is given in Figure 1. The first month after a certifier's visit looks much as the certificate says it does, at 92% fully compliant. The share falls through the first two quarters and then stops falling: the 53-to-104-week band and the 105-to-156-week band differ by one percentage point, which is less than the difference between either of them and the band before. Fitted occupancy reaches within one point of the stationary distribution (0.598, 0.284, 0.118) by week 26.

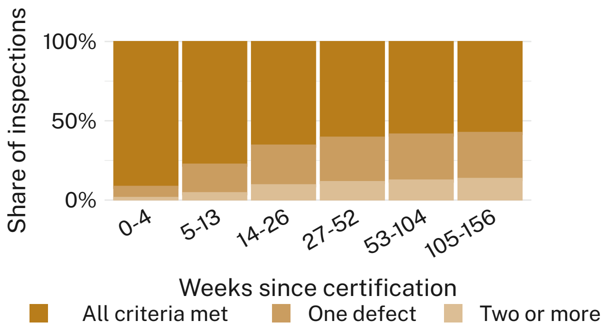

Figure 1: Barrier state at unannounced inspection, by weeks since certification. The composition settles by the second quarter of a 156-week certificate and moves by one percentage point across the remaining two years.

The transition intensities behind that pattern are given in Table 1. Departure from full compliance is common and return is nearly as common: estimated at 0.068 and 0.049 per barrier-week respectively. Progression to the worst state is an order of magnitude rarer than movement between the first two.

Transition	Intensity (per week)	95% CI
$S_1 \to S_2$	0.068	0.055 to 0.084
$S_2 \to S_1$	0.049	0.038 to 0.062
$S_2 \to S_3$	0.021	0.015 to 0.029
$S_3 \to S_2$	0.038	0.027 to 0.053

Table 1: Fitted transition intensities. A certified barrier leaves full compliance at roughly one and a half times the rate at which it returns, and the chain spends most of its time moving between the first two states.

Mean sojourn times follow directly (Figure 2). A barrier holds the state its certificate names for 14.6 weeks on average (95% CI 11.9 to 18.2), against a certificate valid for 156. The single-defect state is briefer still at 9.7 weeks, and the worst state is the most persistent at 20.4 weeks, which is consistent with defects that require a trade rather than a decision.

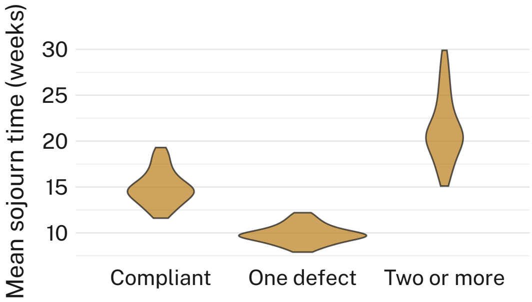

Figure 2: Bootstrap distributions of mean sojourn time by state. The state named on the certificate is the shortest-lived of the three except the one that resolves itself.

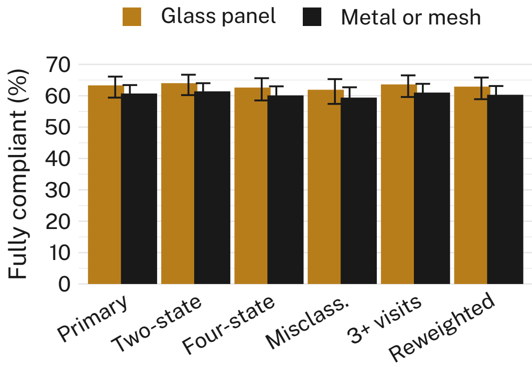

Figure 3: Point prevalence of full compliance under six pre-specified specifications, by barrier material. No specification moves the estimate outside the interval of any other.

The defects are ordinary. Among single-defect observations, 57.9% were an object standing inside the non-climbable zone — most often a garden chair, a planter, or a wheeled bin — 31.4% a gate that closed but did not latch from a partly open position, 7.2% a gap beneath a fence panel, and 3.5% everything else. Of single defects present at one inspection, 84.3% were absent at the next, at a median observed interval of 11 weeks. These are not failures that persist; they are failures that recur.

Household and barrier characteristics explained little. Barrier material, pool age, certifier identity, and whether the household had children under five each moved the estimated S_1 probability by less than four percentage points, and none of the four intervals excluded zero. Time since certification, beyond the first two quarters, was similarly uninformative, which is the substantive point rather than a null to be explained away.

The six pre-specified variants are reported in Figure 3. Estimated point prevalence ranges from 58.9% to 63.5% across all twelve specification-by- material cells, a spread of 4.6 points against confidence intervals averaging 6.8 points wide. Allowing inspector misclassification moves the primary estimate by 1.1 points and widens it; we retain the three-state formulation as primary for completeness.

Discussion

Two numbers describe the same barriers over the same period. One is 98.1% and counts certificates; the other is 61.4% and counts barriers. Both are correctly computed, and the difference between them is not a discrepancy to be reconciled. It is the interval between the certifier's visit and everything else, which the register does not sample and was never designed to.

The finding we did not anticipate is how quickly the population forgets its own certification date. By the end of the second quarter, knowing when a barrier was certified tells an observer almost nothing about its present state, which leaves the certificate carrying information for something under a fifth of its validity. That is a statement about the reporting interval rather than about compliance, and absence of a detectable trend across the remaining two years should not be read as evidence that none exists [21].

We record no criticism of the council, whose scheme is administered exactly as written, nor of the households, most of whom moved the chair back within a fortnight. The University's Indicator Commons has since listed barrier certification among the indicators whose measurement interval and reporting interval are set by different considerations.

Limitations

Participation was voluntary and the consenting third of the invited sample may maintain their barriers more attentively than the rest, which would place the true point prevalence below our estimate rather than above it. Our fieldwork covers one council's register, whose single accreditation pathway and common checklist are what made a consistent verdict possible across 312 dwellings — an unrepresentative feature that is also the reason the classification can be trusted. We did not ask households why an object was where it was, having judged that an account given to an inspector at the gate is a weaker record than the gate.

Conclusion

At a randomly chosen moment, 61.4% of backyard pool barriers holding a current certificate met the standard that certificate attests, against a register figure of 98.1% for the same pools. The fitted chain leaves the certified state after 14.6 weeks on average and reaches its stationary distribution within the first quarter of a three-year certificate, with defects that are small, reversible, and recurrent rather than persistent. Future work will ask what re-inspection interval would make a register's reported rate and its underlying prevalence converge, and whether the same divergence appears in the other certified-once, reported-quarterly schemes a council maintains.

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